Big Data Project Flight Delays Analysis



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Agenda

- Dataset & Big Data challenges
- Architecture: Bronze / Silver / Gold
- Gold layer aggregations
- Power BI dashboard

Dataset & Big Data Challenges

- Large-scale flight data – 91 separate month files (around 7GB)
- Many rows, time-based granularity (date, time, flight level)
- Need for aggregation & optimization

Bronze Layer – Raw Data

- Raw ingestion
- No cleaning or transformations yet
  - Fast ingestion,
  - Reproductible pipeline
  - Data can be reprocessed if needed
- Metadata addition

```
raw_path = "/Volumes/workspace/default/ontime_raw/ontime_data"

df_raw = (
    spark.read
        .option("header", True)
        .option("recursiveFileLookup", "true")
        .option("pathGlobFilter", "*.csv")
        .option("inferSchema", "true")
        .csv(raw_path)
)

print("Rows:", df_raw.count())
df_raw.printSchema()
```

```
df_bronze = (
    df_raw
    .withColumn("_source_file", col("_metadata.file_path"))
    .withColumn("_ingestion_ts", current_timestamp())
)

df_bronze.write \
    .format("delta") \
    .mode("overwrite") \
    .saveAsTable("bronze.ontime_raw")
```

Silver Layer – Cleaning & Structuring

- Filtering & type casting
- Prepared for aggregation
- Data check

```python
df = spark.table("bronze.ontime_raw")

def try_int(c):
    return F.expr(f"try_cast({c} as int)")

def try_double(c):
    return F.expr(f"try_cast({c} as double)")

df_silver = (
    df
    .withColumn("flight_date", F.to_date("FL_DATE", "M/d/yyyy h:mm:ss a"))
    .withColumn("year",  try_int("YEAR"))
    .withColumn("month", try_int("MONTH"))
    .withColumn("day_of_month", try_int("DAY_OF_MONTH"))
    .withColumn("day_of_week",  try_int("DAY_OF_WEEK"))
    .withColumn("dep_hour", (try_int("DEP_TIME")/100).cast("int"))
    .withColumn("dep_delay_new", try_double("DEP_DELAY_NEW"))
    .withColumn("arr_delay_new", try_double("ARR_DELAY_NEW"))
    .withColumn("dep_delay_15", (F.col("dep_delay_new") > 15).cast("int"))
    .withColumn("arr_delay_15", (F.col("arr_delay_new") > 15).cast("int"))
    .withColumn("is_cancelled", try_int("CANCELLED"))
)

df_silver.write.format("delta").mode("overwrite").saveAsTable("silver.ontime_clean")
```

```python
chk = spark.table("silver.ontime_clean")
chk.select(
    F.count("*").alias("rows"),
    F.sum(F.col("year").isNull().cast("int")).alias("year_nulls"),
    F.sum(F.col("month").isNull().cast("int")).alias("month_nulls"),
    F.sum(F.col("dep_hour").isNull().cast("int")).alias("dep_hour_nulls"),
    F.sum(F.col("arr_delay_new").isNull().cast("int")).alias("arr_delay_nulls"),
).show()
```

> See performance (1)

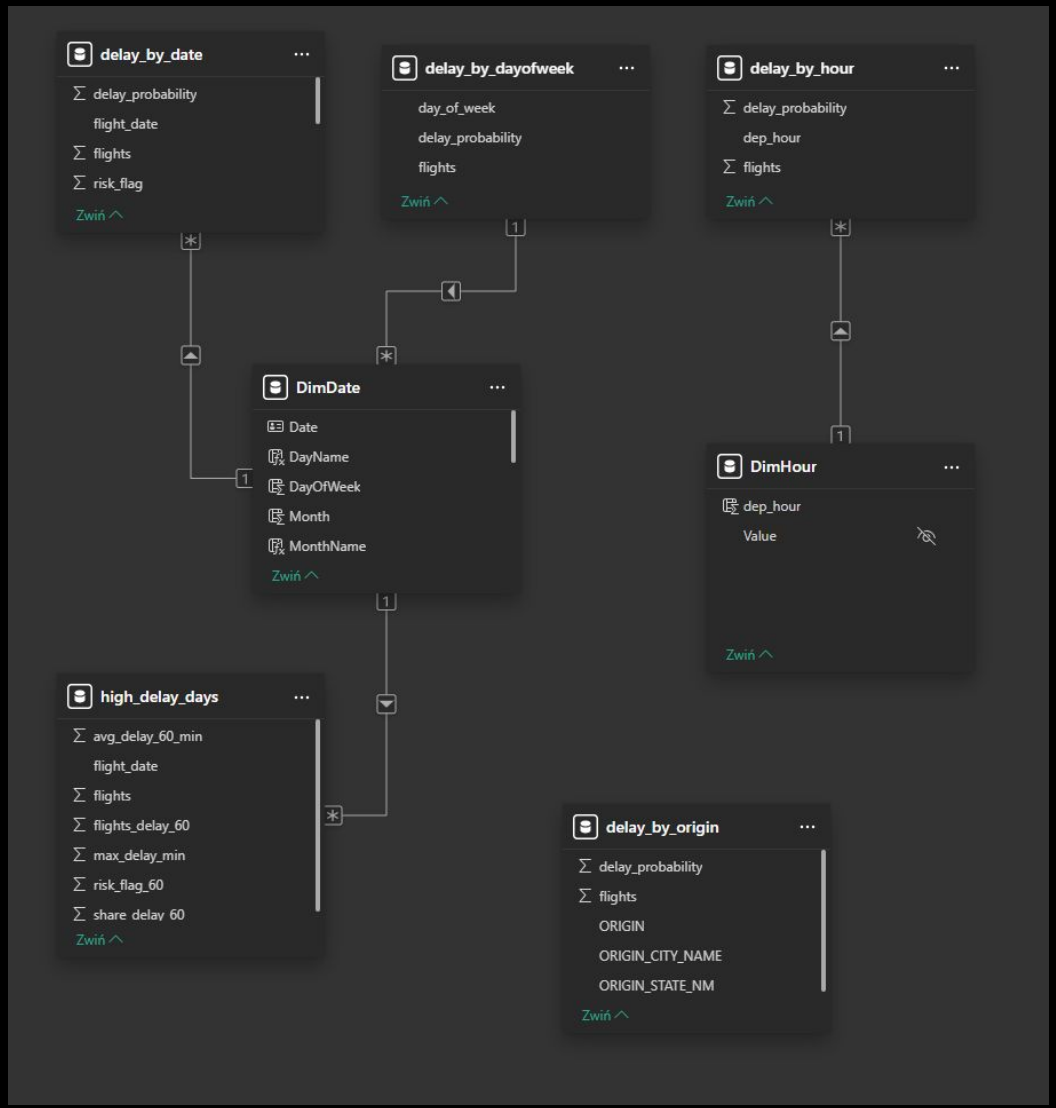
> chk: pyspark.sql.connect.dataframe.DataFrame = [year: integer, QUARTER: string ... 24 more fields]

```
+--------+----------+-----------+--------------+--------------+
|    rows|year_nulls|month_nulls|dep_hour_nulls|arr_delay_nulls|
+--------+----------+-----------+--------------+--------------+
|53619304|         0|          0|       1133805|       1295515|
+--------+----------+-----------+--------------+--------------+
```

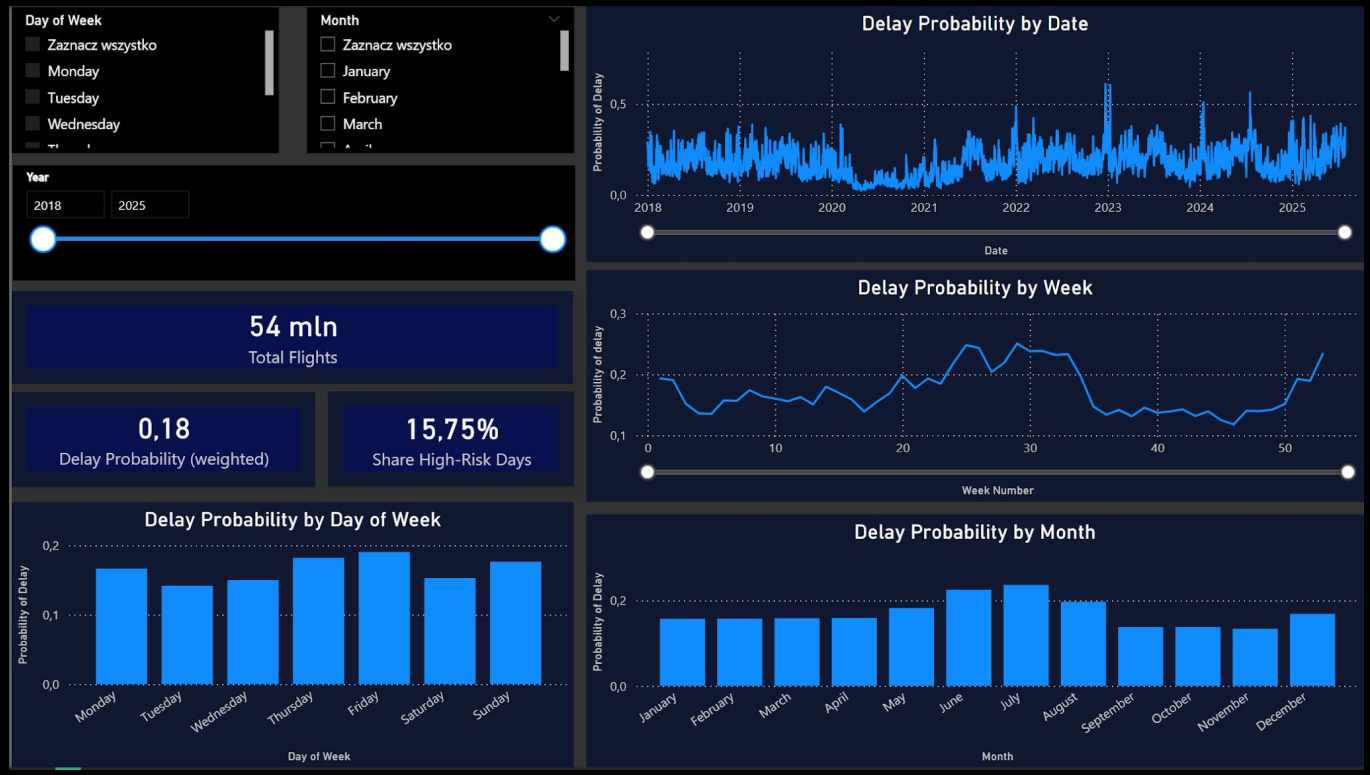
Gold Layer – Aggregations

- Pre-aggregated metrics
  - Creating multiple smaller, aggregated tables for different purposes (delays by hour, by origin, by date, etc.)
  - Calculating some basic metrics (probability, share of flights delayed more than 60 minutes)
  - Creating delay risk flags based on calculated baseline
- Optimized for BI

Power BI – Data Model



Power BI – Visualizations



Why is it useful?

- Based on US data, but patterns can be interpreted globally
- Major holidays like Christmas/ New Year don't differ in dates between Poland and US
- Local patterns easy to notice and exclude - 4th of July
- Same hemisphere -> same seasons -> summer peak season at the same time

Thank you